AGRI FUSION-AI

UAV-Based Soybean Crop Health Prediction Using Machine Learning

Project Report

Submitted in partial fulfillment of the requirements for the course

Fundamentals of AIML

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Title:

AgriFusion: UAV-Based Soybean Crop Health Prediction Using Deep Learning

1. Introduction

With the increasing demand for efficient agricultural practices, early detection of crop diseases has become essential. Traditional methods of monitoring crop health involve manual inspection, which is time-consuming, labor-intensive, and often inaccurate over large farming areas.

This project presents an AI-based solution that uses UAV (drone) imagery and deep learning techniques to automatically predict soybean crop health. By analyzing aerial images, the system helps identify healthy crops and detect disease conditions at an early stage.

The goal of this project is to demonstrate how artificial intelligence can be applied to solve real-world agricultural problems in a practical and scalable manner.

2. Problem Statement

Farmers face difficulty in monitoring crop health across large fields. Manual inspection is not only inefficient but also prone to human error. Delayed detection of diseases can lead to significant crop loss.

This project aims to solve this problem by developing a system that:

- Analyzes UAV images of soybean fields.
- Identifies crop health conditions.
- Provides fast and reliable predictions.

3. Objectives

- To build a deep learning model for crop health classification.
- To use UAV imagery for realistic field analysis.
- To apply transfer learning for efficient model training.
- To enable deployment using the ONNX format.
- To provide a scalable and practical solution for smart farming.

4. Dataset Description

The dataset used in this project is the Soybean UAV Image Dataset obtained from Mendeley Data. It consists of RGB images captured using drones over soybean fields.

The dataset includes four classes:

- Healthy Soybean.

- Soybean Pest Attack.
- Soybean Mosaic.
- Soybean Rust.

The data is divided into training and validation sets. The dataset reflects real-world conditions, including class imbalance, making the problem more challenging and realistic.

5. Methodology

The project follows a structured pipeline:

1. Data Collection
   UAV images are collected and organized into training and validation folders.
2. Data Preprocessing
   Images are resized, normalized, and augmented using techniques such as random cropping and flipping.
3. Model Training
   A pretrained MobileNetV2 model is used. The final classification layer is modified to match the number of classes. The model is trained using cross-entropy loss and the Adam optimizer.
4. Validation
   The model is evaluated on validation data, and the best performing model is saved.
5. Model Export
   The trained model is converted to ONNX format for portability and deployment.
6. Inference
   The ONNX model is used to predict crop health from new input images.

6. Model Selection

MobileNetV2 was chosen due to its efficiency and lightweight architecture. It provides a good balance between accuracy and computational cost, making it suitable for real-time and edge-based applications.

7. Results

The model successfully learns to classify crop conditions based on UAV images. It provides predictions along with confidence scores, enabling reliable identification of crop health.

The use of transfer learning improves performance even with a limited dataset.

8. Challenges Faced

- Handling class imbalance in the dataset.

- Preventing overfitting during training.
- Maintaining consistency between training and inference preprocessing.
- Selecting an appropriate model that balances accuracy and efficiency.

9. Applications

- Crop disease detection
- Smart farming systems
- Agricultural monitoring using drones
- Decision support for farmers

10. Limitations

- Limited dataset size
- Performance may vary under different environmental conditions
- No real-time deployment implemented

11. Future Scope

- Integration with real-time UAV systems
- Expansion to multiple crop types
- Use of advanced deep learning models
- Development of a user-friendly application interface

12. Conclusion

This project demonstrates how artificial intelligence can be effectively applied in agriculture to solve real-world problems. By using UAV imagery and deep learning, the system enables early detection of crop diseases and supports data-driven decision-making.

The project highlights that meaningful solutions do not require excessive complexity, but rather a clear understanding of the problem and practical implementation.